DASH: Detection and Assessment of Systematic Hallucinations of VLMs

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Abstract

Vision-language models (VLMs) are prone to object hallucinations, where they erroneously indicate the presence of certain objects in an image. Existing benchmarks quantify hallucinations using relatively small, labeled datasets. However, this approach is i) insufficient to assess hallucinations that arise in open-world settings, where VLMs are widely used, and ii) inadequate for detecting systematic errors in VLMs. We propose ***DASH*** *(Detection and Assessment of Systematic Hallucinations), an automatic, large-scale pipeline designed to identify systematic hallucinations of VLMs on real-world images in an open-world setting. A key component is DASH-OPT for image-based retrieval, where we optimize over the "natural image manifold" to generate images that mislead the VLM. The output of DASH consists of clusters of real and semantically similar images for which the VLM hallucinates an object. We apply DASH to PaliGemma and two LLaVA-NeXT models across 380 object classes and, in total, find more than* $19k$ *clusters with* $950k$ *images. We study the transfer of the identified systematic hallucinations to other VLMs and show that fine-tuning PaliGemma with the model-specific images obtained with DASH mitigates object hallucinations. Code and data are available at* https://YanNeu.github.io/DASH.

1. Introduction

While vision-language models (VLMs) demonstrate remarkable comprehension of multimodal text and image data, they suffer from object hallucination errors such as incorrectly identifying objects that are not present or describing incorrect relationships or positions of objects within image descriptions [29].

Benchmarks like POPE [24] and AMBER [51] assess the erroneous indication of the presence or absence of an object by a VLM, but are limited by the curated datasets they use, such as MSCOCO [25]. While these benchmarks provide a first assessment of these errors, they have two main issues:



Figure 1. **DASH:** Systematic Hallucinations of PaliGemma-3B.

1. There is no systematic assessment of the types of images for which VLMs hallucinate, making it difficult to determine whether these errors are random or indicative of a systematic issue.
2. The reliance on small, curated datasets like MSCOCO with a small number of object classes fails to accurately reflect the open-world application of VLMs in real-world scenarios. This could potentially lead to overlooking significant problems due to the limited and biased nature of these datasets.

In this paper, we focus on object hallucinations of VLMs where the models respond "yes" to the question "Can you see an *object* in this image?"[1] although the *object* is not actually present. For brevity, we refer to this type of object hallucination as **false-positive-hallucination (FP-hallucination)** throughout the rest of the paper. The opposite case, where the VLM incorrectly answers "no" despite the object's presence, is also of interest but can be checked with existing detection benchmarks. As we demonstrate in this paper, object hallucinations may occur in entirely unexpected contexts (see Fig. 1). Thus, addressing them requires an open-world approach not limited to small benchmark datasets. Instead we argue that assessing and fixing object hallucinations of a VLM requires (i) exploring

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[1]We apply the suffix "Please answer only with yes or no." to ensure valid VLM responses.

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web-scale datasets like ReLAION-5B [39], and (ii) finding model-specific hallucinations for the given VLM. We make the following contributions in this paper:

- We propose DASH (Detection and Assessment of Systematic Hallucinations), a large-scale, fully automated pipeline requiring no human labeling for identifying systematic FP-hallucinations in VLMs by detecting clusters of semantically similar images causing them. It consists of DASH-LLM, text-based retrieval using queries generated by a LLM, and DASH-OPT, image-based retrieval with generated images in ReLAION-5B.
- In DASH-OPT, we propose a method for optimizing the generative process of a latent diffusion model to produce images where the VLM hallucinates an object, while an open-world object detector has low confidence that the object is present.
- We apply DASH-OPT and DASH-LLM to PaliGemma [3] and LLaVA-NeXT (Mistral and Vicuna) [28]. In total, we find more than $19k$ clusters of object hallucinations comprising more than $950k$ images.
- We show that the hallucinations for the found images successfully transfer to seven other VLMs, including one of the top open-weights models, QwenV2-72B [52], on the HuggingFace Open VLM leaderboard [8].
- We propose a new benchmark, DASH-B, to enable a more reliable evaluation of this issue in current VLMs.
- We show that fine-tuning PaliGemma on our large dataset of object hallucinations can help mitigate the problem.

2. Related Work

Hallucinations of VLMs: Benchmarks and Mitigation. An early work to benchmark hallucinations in short image captions of VLMs was CHAIR [41]. A recent study [18] differentiates between type I hallucinations in free-form answers and type II hallucinations in response to factual questions about the image (*e.g.* "Is there a car in this image? Answer yes or no"). POPE [24] benchmarks type II hallucinations; however, being limited to 80 objects in MSCOCO it does not capture the variability in VLM usage. AMBER [51] addresses different kinds of hallucinations including "existential" (object hallucinations). While object hallucinations saturate in their benchmark for VLMs, we demonstrate that a substantial number of systematic object hallucinations persist even in current models such as Qwen2-VL-72B [52] and Llama 3.2-VL-11B [9]. Other recent benchmarks such as HALO-QUEST [53] (type II) or MMVP [50] (type I and II) cover multiple modalities but are too small to provide sufficient statistics on specific VLM errors.

The literature on mitigation of hallucinations is limited. Visual contrastive decoding [21] is a training-free method contrasting original and noisy images, robust instruction tuning [26], LURE [62] is a post-hoc method aimed at reducing hallucinations, and M-HalDetect [13] provides a detection (type I) and mitigation technique.

Spurious Correlations/Hallucinations in Image Classification: Spurious correlations in image-based data [5], such as a cow on the beach not being recognized due to the absence of the spurious feature (e.g., grass), are known issues in image classification. In [34, 47], spurious features are identified on ImageNet, and [34] proposes "Spurious ImageNet", a benchmark for hallucinations in ImageNet classifiers (classifier predicts presence of class despite not being present in the image) for 100 ImageNet classes.

Debugging of ML Models using Guided Image Generation. Zhang et al. [61] utilize stable diffusion to generate images with variations in background, material, and texture to deceive models like CLIP in zero-shot classification, showing partial transfer to VQA-type tasks in miniGPT4 and LLaVA-1.5. Similarly, Metzen et al. [32] identify errors in image classifiers on rare subgroups by evaluating generated images with variations in color, size, background, and weather. DiG-IN [1] is a debugging technique that uses optimization-guided image generation, for instance, to find images that maximize prediction differences for a certain class between two classifiers. [1] employ this approach to systematically identify subgroups that CLIP misclassifies.

3. DASH: Detection and Assessment of Systematic Hallucinations

The goal of DASH is to identify systematic FP-hallucinations by detecting clusters of semantically similar images that trigger them (see Fig. 1 or 3 for examples found by DASH). This type of object hallucination is increasingly relevant as AI agents begin to automatically process image data. We tackle this challenge by searching over ReLAION-5B, avoiding the limitations of smaller datasets that cover only few object classes compared to the unrestricted application of VLMs. Importantly, an object category's presence in a dataset does not imply that only objects from this dataset could lead the VLM to hallucinate. Thus, benchmarks such as POPE [24] or AMBER [51] underestimate the problem of FP-hallucinations, while exhaustive testing of each image-object combination, as in [18], is infeasible for ReLAION-5B.

DASH is a fully automatic pipeline, designed to require no human labels. While this carries the risk of errors—which we test in our evaluation—it enables DASH to produce large-scale datasets that can be used for evaluation and fine-tuning of VLMs. To effectively detect systematic hallucinations, we propose two approaches: DASH-OPT, based on images specifically generated to deceive the VLM, and DASH-LLM, based on LLM-generated queries. Subsequently, these image and text queries are used in an exploration phase to find real images in ReLAION-5B which trigger FP-hallucinations. In an exploitation step, we verify that these are not isolated errors by identifying semantically

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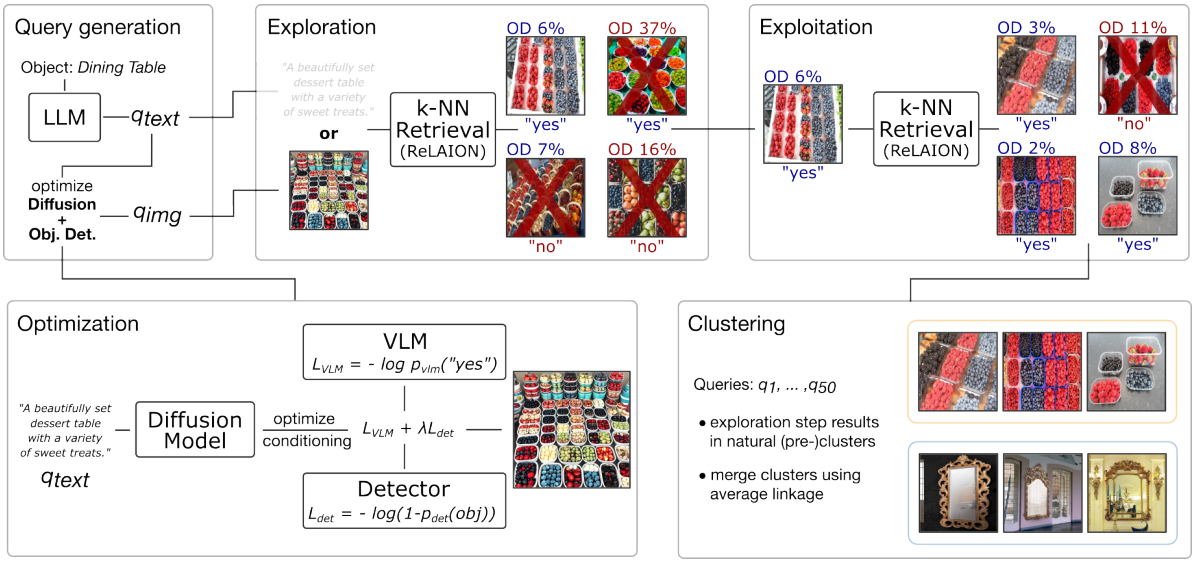


Figure 2. **DASH:** Given an object class, e.g. dining table, we generate text-based queries with **DASH-LLM** or image-based queries with **DASH-OPT**. **Optimization:** we optimize the latent variables of a diffusion process to generate an image which yields "yes" for the VLM ("Can you see a dining table in this image?") and at the same time the object detector states that no "dining table" is present in the image. **Exploration:** the text and image queries are used for kNN-retrieval using CLIP similarity on ReLaion-5B. **Exploitation:** for successful images (VLM "yes", object detector "no") of the exploration phase we retrieve novel images via kNN-retrieval to check if the hallucination transfers to semantically similar images. **Clustering:** Finally, we cluster successful images of the exploitation step into semantically similar clusters of hallucinations of the VLM.

similar images that consistently deceive the VLM without containing the object. Finally, as different queries may yield overlapping image groups, we cluster the results (see Fig. 5 for samples from all clusters for the object "dam" found by DASH). We summarize the workflow of DASH in Fig. 2.

3.1. DASH-LLM

FP-hallucinations often arise from associations between the target *object* and other objects present in an image. This can result from co-occurrence at the image level (i.e., objects that are frequently photographed together) or co-occurrence in texts. We leverage the fact that large language models (LLMs) are trained on extensive text corpora and are well-suited for generating lists of candidate prompts. Specifically, we use Llama 3.1-70B with a carefully crafted system prompt to generate 50 text prompts, $q_1, \ldots, q_{50}$, for each object. We ask the model to "create prompts that could lead an AI model to falsely recognize the object due to the presence of spurious features, even though the object itself is not present in the images" (see Appendix B). The LLM is instructed to avoid mentioning the object itself or any of its parts in the prompts and to avoid repeating queries.

While this approach to generating text queries is simple, it proves to be quite effective. However, it has two limitations: first, it is entirely agnostic to the specific VLM being used. Second, even if the generated text queries are conceptually capable of causing the VLM to hallucinate, our CLIP-similarity-based retrieval on ReLAION-5B (see Sec. 3.3), may fail to retrieve the appropriate images. We address these limitations of DASH-LLM by using image queries generated through DASH-OPT, which are directly optimized to induce hallucinations in the VLM.

3.2. DASH-OPT

The goal of DASH-OPT is to generate images that cause FP-hallucinations in the VLM. As our pipeline is supposed to be fully automatic, we require an alternative to human verification to determine whether the image contains the object. For this purpose, we employ an open-world object detector—in this case, OWLv2 [33]—with a very low detection threshold. This minimizes the chance of missing the object when it is actually in the image. However, some images may be incorrectly flagged as containing the object. We discuss this trade-off in more detail in Appendix G.

Typically, optimizing in pixel space results in adversarial samples; however, previous work [1] has shown that meaningful solutions in the natural image manifold can be obtained by optimizing the input variables of a diffusion process. While prior works [1, 43] rely on multi-step diffusion processes—which are computationally intensive—we utilize recent advances in diffusion distillation [40] and use a single-step diffusion process, significantly reducing computational cost. We denote by $q(C)$ the output image of the diffusion process based on its conditioning C (see Appendix C for details). Below, we describe the two objectives we optimize to obtain the desired query image $q(C)$.

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We denote the text query "Can you see a OBJ in this image?" by qstnOBJ for the considered object class OBJ. Since "yes" corresponds to a single token, we can directly optimize the cross-entropy loss for the standard next-token prediction of "yes" in the VLM based on the image $q(C)$:

$$L_{\text{vlm}}(C) = -\log p_{\text{vlm}}\left(\text{"yes"} \mid q(C), \text{qstnOBJ}\right). \quad (1)$$

Similarly, we propose a loss function to penalize the object detector's confidence in detecting the object class OBJ in the generated query image $q(C)$:

$$L_{\text{det}}(C) = -\log\left(1 - p_{\text{det}}\left(\text{OBJ} \mid q(C)\right)\right), \quad (2)$$

where $p_{\text{det}}\left(\text{OBJ} \mid q(C)\right)$ is the object detector's confidence that the generated query image $q(C)$ contains the class OBJ.

The final optimization problem for the conditioning C generating the image $q(C)$ is then given as:

$$\min_{C} \; L_{\text{vlm}}(C) + L_{\text{det}}(C) \quad (3)$$

This objective aims to make the VLM hallucinate the object in the resulting image $q(C)$, while the detection loss ensures that the optimization process does not simply insert the object into the image.

In practice, we initialize the optimization of C in the diffusion process using real text prompts encoded by the diffusion model's text encoder. We start with the same 50 LLM-generated prompts from DASH-LLM, creating 50 image queries $q_1, \ldots, q_{50}$. However, our optimization process steers the resulting image $q(C)$ away from images that either fail to fool the VLM or actually contain the object. Importantly, while the initial text prompts are derived purely from the LLM and are independent of the VLM, the image queries are directly optimized for the VLM, allowing us to uncover model-specific issues.

Notably, the final queries $q(C)$ often differ substantially from the initial text prompt outputs (see discussion in Sec. 4.1), leading DASH-OPT to discover a larger variety of clusters compared to DASH-LLM as well as achieving a higher success rate in producing images that mislead the VLM, as shown in Tab. 1. See Fig. 3 for examples of generated queries $q(C)$ for the objects tench, leopard, piano.

3.3. Exploration, Exploitation and Clustering

Given the 50 text queries generated with DASH-LLM respectively 50 image queries generated with DASH-OPT (denoted as q_{text} and q_{img} in the overview of the pipeline in Fig. 2), our goal is to find real images for which the VLM hallucinates the object OBJ even though it is not visible in the image. Our pipeline is entirely source-data-free, requiring only object labels and no human supervision in subsequent stages, making it easily scalable to large datasets.

In the **exploration phase**, let $q_1, \ldots, q_{50}$ denote the 50 text or image queries generated with DASH-LLM or DASH-OPT for a given object. For each query, we retrieve 20 images from ReLAION-5B using the fast CLIP kNN-index of [2, 7], yielding a total of 1,000 images per object.

We then filter out all images where the object detector flags that the object is contained in the image. We use a very conservative threshold, as discussed above, to ensure that the object is not mistakenly contained in the image. While this reduces our success rate, ensuring that the object is not present is crucial for the pipeline; otherwise, the VLM does not hallucinate when replying with "yes". We evaluate our automatic object-detector-based approach against a human baseline in Section 4.1. Among the remaining images, we filter out all images that fail to trigger a FP-hallucinationin the VLM. We call this stage **exploration**, as we explore whether retrieval based on our text and image queries leads to successful hallucination on real images. Furthermore, the exploration phase aims to generate a diverse set of candidates for potential systematic hallucinations.

While the goal of the exploration phase is to provide a diverse set of candidates, the **exploitation phase** aims to achieve a high VLM acceptance rate and reveal systematic hallucinations. During this phase, we retrieve 50 images for each image candidate from the exploration phase, again using kNN-retrieval on ReLAION-5B. However, note that we are effectively retrieving a larger set of nearest neighbors as potential candidates since we filter out near duplicates using the perceptual metric DreamSim [11] with a threshold of 0.9. This is necessary, as ReLAION-5B contain identical images or slight variations in resolution, viewpoint, or cropping. Our goal is to find semantically similar images in the sense of similar image composition and object content but not just small variations. While the exploration phase can use either text or image queries, the exploitation phase always uses image queries derived from the exploration phase. This is particularly useful since it results in a natural clustering of semantically similar images. To make this even more explicit, we further merge several of these preclusters, *i.e.* images that are neighbors of the same source image, using agglomerative clustering with average linkage based on DreamSim distances. Note that we overload notation by using DASH-LLM and DASH-OPT to refer to the results of the full pipeline, not just the initial query generation. All stages of the pipeline are summarized in Fig. 2.

4. Experiments

For our experiments, we use a total of 380 object categories. Specifically, we include the 80 classes from COCO [25], 100 randomly selected object categories from Objects365 [45] and 100 classes from ImageNet which have been used in Spurious ImageNet [34]. Additionally, we sort the objects in OpenImages [20] based on their occurrence frequency and create four subsets, each containing 25 objects. The first subset corresponds to the most com-

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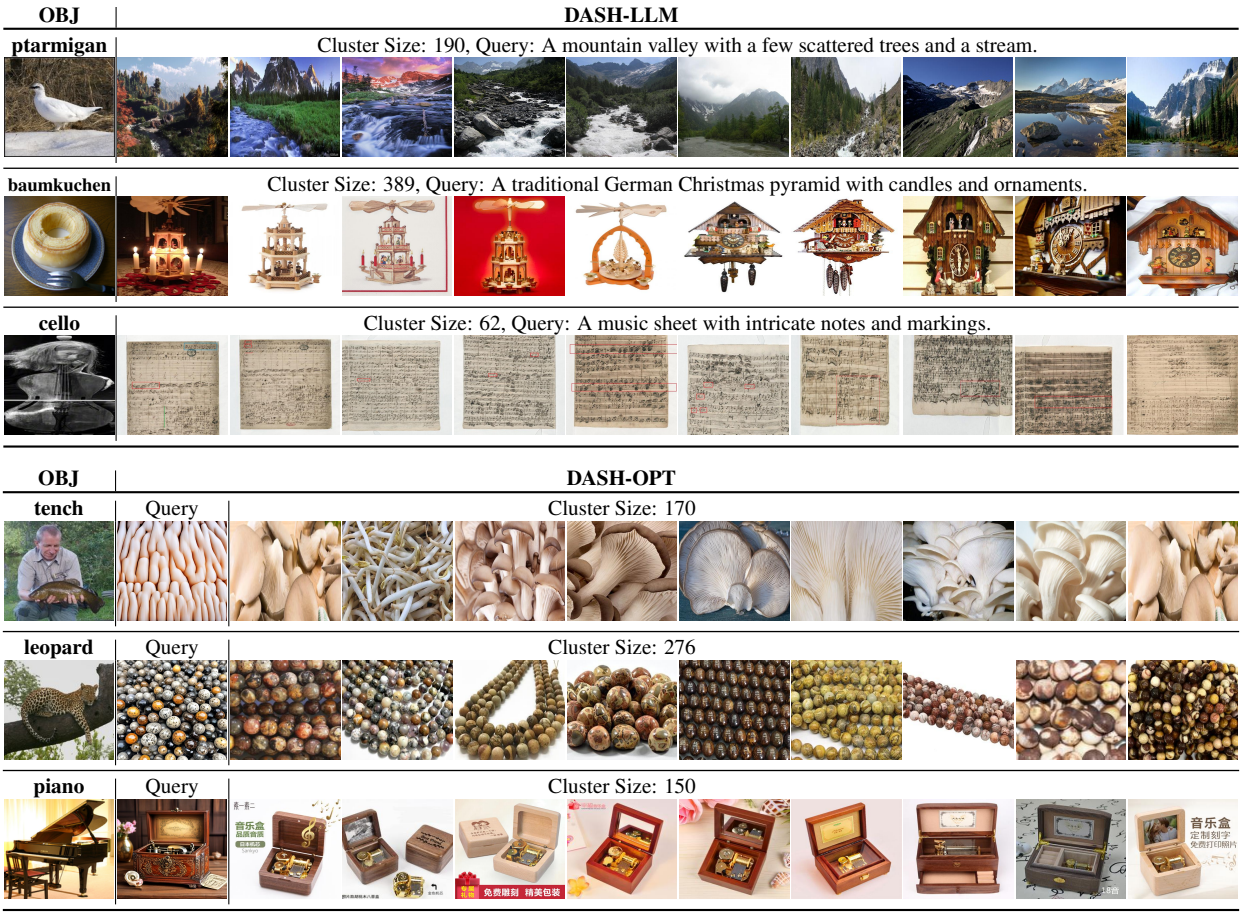


Figure 3. **Examples of systematic FP-hallucination clusters found by DASH for PaliGemma:** We present six hallucination clusters, each for a different object—three identified by DASH-LLM and three by DASH-OPT. For each cluster, we show a sample of images and the total number of images. **For each of these images, PaliGemma answers "yes" to "Can you see a OBJ in this image?"** while the object detector reports a confidence below 0.1. None of the images actually contain the object. We also provide the text (DASH-LLM) and image queries (DASH-OPT) used for retrieval during exploration for the majority of the cluster.

mon objects in OpenImages. The second and third correspond to objects around the 10% quantile and the median in occurrence frequency, respectively. The last subset contains the least common objects, some without a single occurrence. We use these subsets to assess the influence of an object's occurence frequency on the hallucination rate in App. E. For our retrieval pipeline described in Sec. 3, we use PaliGemma [3] and LLaVA-NeXT [28] in the Vicuna [36] and Mistral [16] variants as VLMs. As the large-scale image dataset, we utilize ReLAION-5B [39, 44] with a CLIP index for fast kNN retrieval [2, 7]. See App. L for an experiment on the reverse task: VLM responds "No" although the object is visible.

4.1. Results of DASH

We report the results of DASH in Tab. 1. DASH-LLM finds a total of 99.3K images for PaliGemma, 162.4K images for LLaVA-NeXT Vicuna, and 78.5K images for LLaVA-NeXT Mistral, while DASH-OPT obtains 221.7K, 252.0K, and 133.8K. In addition to more images, DASH-OPT yields a larger number of clusters: 3895/4632/3229 compared to 1892/3632/2001 for DASH-LLM.

Source Model	DASH Variant	Total Images	Total Clusters	Avg Clstr per Object	Avg Imgs per Clstr
PaliG	LLM	99.3K	1892	5.0	52.5
	OPT	221.7K	3895	10.3	56.9
LN Vic	LLM	162.4K	3632	9.6	44.7
	OPT	252.0K	4632	12.2	54.4
LN Mis	LLM	78.5K	2001	5.3	39.3
	OPT	133.8K	3229	8.5	41.5

Table 1. Retrieval results for DASH-LLM and DASH-OPT across PaliGemma, LLaVA-NeXT Vicuna, and LLaVA-NeXT Mistral, accumulated over the 380 object categories from all datasets.

Some of the clusters for both approaches are shown in

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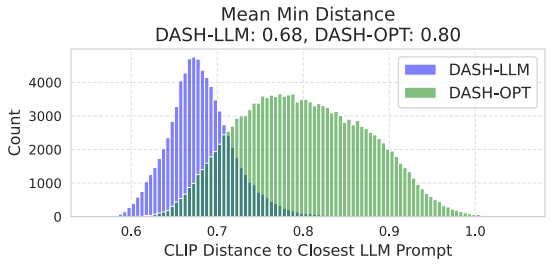


Figure 4. Histogram illustrating the minimum embedding distance from success images to the nearest LLM prompt for DASH-LLM and DASH-OPT. While both methods use these LLM prompts in their exploration stage, the image-based method is able to find unexpected hallucinations far away from the initial LLM prompts.

Fig. 3 (and App. F) along with one of the corresponding text or image queries, respectively. Considering the clusters found by DASH-LLM, the associations between objects and hallucinations are relatively clear and closely correspond to the text queries: a water cannon is often present in images of fireboats; the mountainsides are a natural habitat of ptarmigans; Baumkuchen is part of German Christmas traditions; and music sheets are linked to musical instruments like cellos. On the other hand, the results of DASH-OPT indicate that it can uncover hallucinations in a completely different context by moving away from the initial text queries during optimization. For these subgroups, potential causes are less obvious, e.g. at first sight, images of beads do not seem to be related in any way to the animal leopard. However, there exist so-called "leopard beads", i.e. beads showing a leopard pattern. While the beads in our cluster are not of this type, their existence might play the role of a confounder.

We further validate the larger exploration range of DASH-OPT over DASH-LLM in Fig. 4, where we illustrate the CLIP distance between the image embedding and the text embedding of the closest original text prompt. While both LLM and OPT use the same text prompts, the additional optimization of the OPT variant allows it to find images that are further away. This can also be seen in Fig. 5, where we visualize all clusters detected for the object "dam" using LLaVA-NeXT Vicuna. DASH-LLM results in 4 clusters with a total of 84 images, while DASH-OPT finds 10 clusters comprising 186 images. The clusters resulting from LLM queries represent natural waterfalls and water surfaces and are also detected using OPT queries. However, beyond those, DASH-OPT also surfaces a range of more diverse and unexpected patterns, including different colors (leaves/sunsets) and distribution shifts (comic frogs/comic dragons).

Advantage of a larger retrieval pool: In Fig. 18, we compare hallucinations found by DASH to their nearest neighbors in the curated dataset, which also contains the object. Note that, first, some of the objects (*e.g.*, rubber boot) or types of images (*e.g.*, maps) are not contained in the smaller

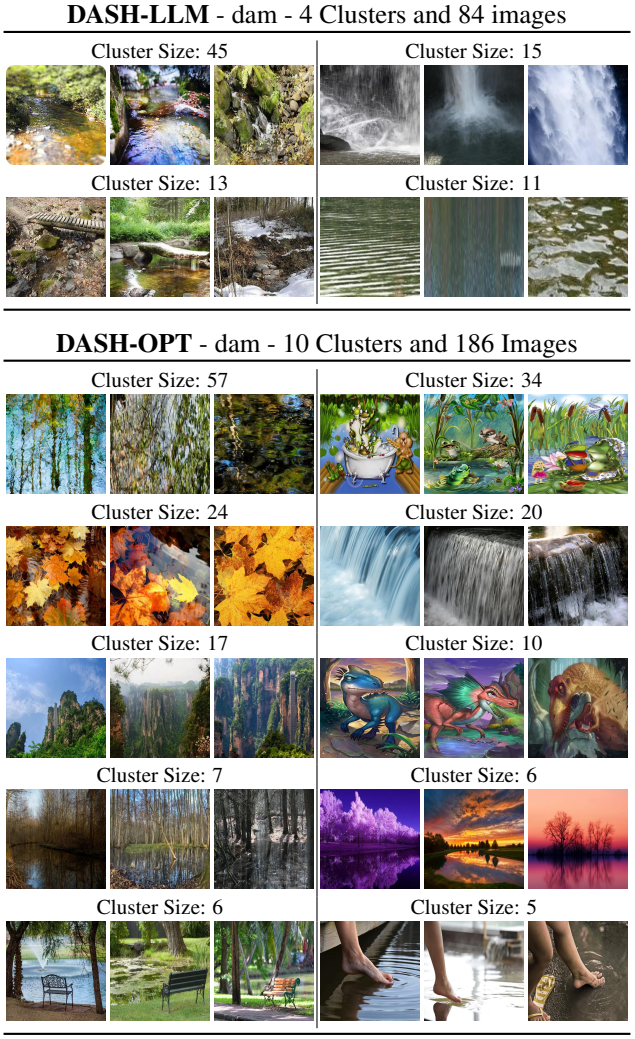


Figure 5. All clusters for DASH-LLM and DASH-OPT for the object "dam" using LLaVA-NeXT Vicuna. DASH-OPT identifies a larger total number of clusters and images, capturing a broader diversity of visuals. This demonstrates that DASH-OPT can uncover unexpected systematic hallucination patterns, such as cartoon frogs and dinosaurs, orange leaves, bare feet, or a park bench, whereas DASH-LLM tends to highlight failure modes more directly linked to the object, such as water associated with "dam".

reference datasets. Second, the most similar images that are contained might not be specific enough to fool the VLM, even when they are similar in object, color, and composition. For example, while Objects365 contains trees, only "Baobab trees" are detected as "sausage". This emphasizes the necessity of a large dataset for searching hallucinations.

Human Verification: To validate the object detector in DASH, we use human supervision to verify the absence of the object. In particular, for all 380 objects found for PaliGemma with DASH-OPT, we randomly select 10 images. We then label them as "yes" if the object is visible, "no" if it is absent, and "ambiguous" for corner cases. For

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example, it can be difficult to assess whether a close-up photo of a flat surface is a "dessert table" or a "coffee table". Across all images, we find that 5.2% contain the object and 7.8% are ambiguous. For comparison, we perform the same annotation on the corresponding subset of POPE, i.e. image/question pairs where PaliGemma responds "yes", but the COCO ground truth indicates "no". Among these 137 alleged false positives, 25.5% contain the object, and 22.6% are marked as ambiguous (examples in App. H). This demonstrates that our conservative threshold for the object detector yields a reliable automatic pipeline with less errors compared to POPE.

Transfer across prompts: We check the influence of the type of the question on the evaluation of the FP-hallucinations found by DASH by testing 11 prompts ("Is OBJ in the image?", "Does this image contain a OBJ?", ...) on PaliGemma, LN Vicuna, and LN Mistral (results in App. M). LN Vicuna has an average "yes"-rate of $82.3\% \pm 6.7\%$, similarly LN Mistral $78.9\% \pm 7.0\%$ showing that the prompt has only a minor influence. For PaliGemma, which was trained on this task on OpenImages, the "yes"-rate drops to $71.6\% \pm 18.7\%$ with higher variance as prompts similar to the training prompt ("Is OBJ in the image?", 31% "yes") show a lower transfer.

4.2. Transfer to unseen VLMs

DASH searches specifically for images causing hallucinations for a target model. In this section, we investigate how the found systematic hallucinations transfer to unseen VLMs. Tab. 2 reports the transfer rate, defined as the proportion of images found by DASH (for PaliGemma, LLaVA-NeXT Vicuna, and LLaVA-NeXT Mistral) that trigger a FP-hallucination in the model. We use these transfer rates to quantify the impact of the LLM backbone, vision encoder, and model scale on FP-hallucinations. In addition, to quantify the detection-hallucination trade-off, we evaluate the true positive rate on a subset of images from COCO, Object365, and ImageNet showing the object (see App. I.2) which we call TPR-ICO.

Influence of LLM backbone: We examine the LLM backbone's influence on the model's vulnerability to systematic hallucinations by considering three versions of LLaVA-NeXT, based on the LLMs Vicuna, Mistral, and Llama. Apart from the LLM, these three share the same vision backbone, architecture, and training procedure. Thus, differences in their robustness to hallucinations can be attributed to the LLM. On images found for PaliGemma, the Vicuna variant has the highest transfer rates (DASH-LLM 49% and DASH-OPT 43%), followed by Mistral (31% and 23%) and Llama (26% and 22%). The TPR-ICO indicates (Vicuna 87%, Mistral 83%, Llama 81%) that the models with higher transfer rates in general reply with "yes" more frequently. Overall, one can conclude that the LLM backbone significantly impacts the vulnerability to FP-hallucinations.



Figure 6. **Object hallucination benchmark DASH-B:** examples from the negative set of DASH-B (images and object label) where GPT-4o-mini, the best scoring model on DASH-B (see Tab. J) hallucinates the object even though it is not present in the image.

Influence of vision encoder: Similar to the LLM backbone, we compare three Prismatic [17] models that share the same LLM but use different ViT-L [6] vision encoders, namely CLIP [38] and SigLIP [60]. On average, CLIP results in a higher transfer rate of 60% while the SigLIP variant is only fooled by 43% of the images, showing that the choice of the vision encoder has a major impact. However, smaller hallucination transfer rates also come with a smaller TPR-ICO, *i.e.* SigLIP (77%) is less likely to respond with "yes", even on data containing the object, than CLIP (83%).

Model size: We evaluate Qwen2-VL [52] both with Qwen2-7B [58] and 72B as language backbone. Overall, the size of the used LLM seems to have only a small effect on the VLM's vulnerability to hallucinations (18% to 19%) but as both models have the same TPR-ICO of 85%, model scaling can improve the detection-hallucination tradeoff.

4.3. DASH-B: Object Hallucination Benchmark

As the popular POPE [24] benchmark for object hallucinations seems saturated and contains significant label noise, a novel hard object hallucination benchmark is needed to measure further progress. For this purpose we use all images found by DASH which transfer to both Qwen2-VL and Llama 3.2-VL (see App. J details). We verify those images for a selection of 70 objects by consensus of two human labelers and limit the minimal and maximal number of images to 3 and 50, respectively. This results in 1341 images, see Fig. 6 for examples. We use this as our set of negatives and add the same amount of images containing the objects. As performance measure, we use accuracy over all 2642 images. Tab. 3, contains the results for four models not used in the generation of DASH-B. Given the label noise (see App. H), the high true negative rates (TNR) on POPE

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VLM Type		PaliGemma	—LLaVA-NeXT—			—Prismatic—		—Qwen2-VL—		Llama 3.2-VL
Vision Encoder		SigLIP	CLIP	CLIP	CLIP	CLIP	SigLIP	Custom	Custom	Custom
LLM		Gemma	Vicuna	Mistral	Llama 3.0	Vicuna	Vicuna	Qwen2-7B	Qwen2-72B	Llama 3.1
PaliG	LLM	-	0.49	0.31	0.26	0.60	0.44	0.18	0.18	0.09
	OPT	-	0.43	0.23	0.22	0.49	0.34	0.17	0.15	0.10
LN Vic	LLM	0.34	-	0.39	0.30	0.64	0.44	0.18	0.16	0.10
	OPT	0.30	-	0.33	0.27	0.59	0.38	0.15	0.13	0.09
LN Mis	LLM	0.39	0.67	-	0.42	0.67	0.51	0.25	0.23	0.14
	OPT	0.35	0.66	-	0.41	0.63	0.46	0.24	0.21	0.15
Average transfer		0.35	0.56	0.31	0.31	0.60	0.43	0.19	0.18	0.11
TPR-ICO		0.81	0.87	0.83	0.81	0.83	0.77	0.85	0.85	0.80

Table 2. **Transfer of DASH images (rows) to different VLMs (columns)** Different LLM backbones (LLaVA-NeXT) and different vision encoders (Prismatic) have a significant impact on the vulnerability to FP-hallucinations, but the LLM size (Qwen2-VL) shows only a small effect. TPR-ICO is the true positive rate calculated on ground-truth validation data from ImageNet, COCO, and Objects365 corresponding to the employed object categories.

Models	PaliG2-3B	Ovis2-8B	LLaVa-OneVision	4o-mini
DASH-B Acc.	68.9%	71.4%	75.1%	86.3%
DASH-B TNR	40.9%	44.8%	60.1%	76.7%
POPE TNR	97.3%	94.9%	95.8%	-[2]

Table 3. **Object Hallucination Benchmark DASH-B:** Compared to POPE, DASH-B is not saturated as demonstrated by the substantially lower true negative rates (TNR). More results in Tab. 6.

	DASH-B	Amber Ex.	R-Bench	POPE	Caption	VQA
PaliG	56.4%	93.2%	79.9%	**87.2%**	**101.0**	**70.4%**
+ft	**68.0%**	**95.4%**	**80.2%**	86.4%	99.2	69.5%

Table 4. **Fine-tuning on DASH:** Our fine-tuning strategy improves performance on DASH-B, Amber Existence and R-Bench.

($96.0\% \pm 1.2\%$) suggest that the benchmark is saturated and provides limited insight regarding FP-hallucinations. In contrast, the models exhibit a significantly lower TNR on DASH-B ($48.6\% \pm 10.1\%$). We show results on DASH-B for more models in Tab. J.

4.4. Fine-Tuning on DASH

We test the usage of the images found by DASH for fine-tuning PaliGemma in a small proof-of-concept (details in App. K): First, we merge the results of DASH-LLM and DASH-OPT for PaliGemma and ensure that none of the images of DASH-B are part of this train set. Per object, we sample 200 random images from this set and additionally 400 positive samples, i.e. images that contain the object. During fine-tuning, the models is trained to output "no" on DASH images and predict "yes" on the positive samples (see App. K for more details on the fine-tuning setup).

Tab. 4 contains the results after finetuning: Accuracy is significantly improved on DASH-B ($+11.6\%$) due to an increase in TNR and TPR (see App. K) and also increases on Amber Existence [51] ($+2.2\%$) as well as R-Bench [57] ($+0.3\%$). However, we observe a small decrease of accuracy on POPE. We also evaluate two captioning tasks (Flickr30k, COCO) and two VQA tasks (TextVQA, VQAv2) and report the averaged results. The minor performance drop (captioning -1.8, VQA -0.9%) is expected as we are fine-tuning the model on a different task. Generally, the DASH data should be integrated into a curriculum learning scheme, as one of several tasks, which is out of scope for this work.

[2]In our POPE evaluation, GPT-4o-mini only provided valid replies for 77.5% of the images and achieved a TNR of 92.7% among those.

5. Conclusion and Limitations

We have demonstrated that DASH is an effective, automatic pipeline for identifying systematic hallucinations in VLMs. Contrary to the belief that object hallucinations are no longer an issue, we find that they persist extensively when using VLMs in an open world scenario. To address this, we propose DASH-B for a more rigorous assessment of these errors. Our initial experiments on mitigation suggest that DASH could be a valuable addition to the training pipeline of future VLMs.

Limitations: we note that achieving exhaustive coverage of all systematic hallucinations with DASH is not possible, as this would require a fully exhaustive approach. Even with such an approach, and despite ReLAION-5B providing significant coverage of natural images, some images remain underrepresented. As a result, even when we identify an image where the VLM hallucinates, there may not be enough semantically similar images in ReLAION-5B to consider it a systematic hallucination. For the most advanced VLMs, our conservative threshold for the object detector could pose a limitation, potentially.

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